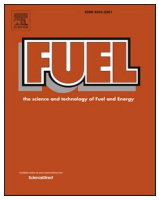
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Full Length Article



Importance of digester's headspace geometry for the efficient H_2S removal through microaeration; experimental and simulation study

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ABSTRACT

Microaeration for biogas desulfurization occurs mainly in the headspace of anaerobic digesters where the biofilm growth is mainly reported; however, no study has been focused on the effect of headspace geometry on microaeration performance. This research evaluates the effect of headspace geometry on biogas flow, subsequent biofilm growth, and the effect of these parameters on microaeration efficiency. This effect was studied in an anaerobic digester with three different headspace configurations. Also, the computational fluid dynamics (CFD) model explained the experimental results. The CFD model developed described the biogas flow in all headspace configurations. It was proven that the modifications of headspace geometry altered biogas flow and avoided biofilm growth on the reactor walls leading to the reactor's protection from the adverse effects of biofilm growth. The mathematical model developed in this work can help design microaerobic digesters or find the best way to implement microaeration into the digesters already built.

1. Introduction

Hydrogen sulfide (H_2S) is an inevitable product of the anaerobic treatment of sulfur-rich substrates [1]. Sulfur-rich substrates are, e.g. sludges from urban wastewater treatment plants (WWTP), agricultural resources, or animal waste. The amount of total sulfur in g/kg (sulfur/total solids) in these substrates is, e.g. 4.5 (primary sludge), 8.6 (activated sludge), 4.0 (camelina crops), 8.0 (pig slurry) [2]. Poisonous H_2S causes concrete and steel corrosion in anaerobic digesters leading to the shortening of the technology's lifetime. Also, H_2S in biogas acidifies engine oil in cogeneration units and can disqualify biogas from upgrading to biomethane [3]. Therefore, it is of great importance to remove H_2S from biogas before its further use.

Several methods for biogas desulfurization are already used in industry, including, e.g. biotrickling filters, biofilters, or bioscrubbers. Detailed information and comparison of these methods can be found in Andreides et al. [4]. In the past decade, microaeration, a bio-based method, has garnered significant attention owing to its high efficiency and relative simplicity [5]. Despite its widespread application in full-scale installations, the theoretical description of this technique fundamental bases have not been adequately elucidated. To address this gap, the present study utilizes mathematical modeling to enhance our current understanding of its fundamental principles.

In microaeration, H_2S is biologically oxidized to harmless elemental sulfur by sulfide oxidizing bacteria (SOB). The growth of SOB is supported by controlled dosing of air or oxygen into the anaerobic digesters [6]. Next to desulfurization, dosing of limited air/oxygen into the liquid phase enhances the hydrolysis of organic material and increases methane production [7]. ORP-controlled based oxygenation can keep the level of sulfide concentration in the liquid phase [8]. The impact of several factors on microaeration efficiency has been documented, including air dose (e.g. Díaz et al. [9], biogas residence time in the headspace [10], pH in the liquid phase (e.g. Krayzelova et al. [11], and the position of oxygen dosing point (e.g. Díaz et al. [12]. The results vary throughout the literature with no apparent explanation. There is a presumption that the microaeration strategy decision could be based on the headspace geometry and position of the oxygen dosing point.

It was proven that regardless of whether oxygen is dosed into the liquid or gas phase, desulfurization occurs mainly in the digester's headspace[13]. When there is no headspace in the digester, desulfurization does not take place since SOB are usually present in the biofilm covering the walls of the headspace where H_2S and oxygen is available [3,12]. When oxygen is provided into the liquid, microaeration usually is not efficient enough, since oxygen is used for oxidation of organic compounds and other processes. The increase in oxygen supply, however, could disrupt the anaerobic conditions. I.e., providing sufficient

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surface area for the growth of SOB biofilm is vital for assuring efficient removal of H_2S from biogas. This biofilm can, however, cause clogging of biogas pipes and destruction of the structural materials [3,5,13]. According to our best knowledge, no clear protection strategy against harmful biofilm effects is defined in the literature.

Headspace geometry can affect the pattern of biogas flow and the efficiency of microaeration. Also, it can direct biofilm growth to locations where harmful effects will be mitigated. However, the effect of the headspace geometry on biogas and/or oxygen flow and subsequent biofilm growth and microaeration efficiency has not been evaluated yet. Moreover, headspace geometry is usually not even mentioned in the research papers dealing with microaeration efficiency. Nevertheless, there is an assumption that if the biofilm growth were transferred from the reactor walls to a properly designed "sacrificial object", the reactor walls would remain protected, thereby extending the technology's lifespan.

In a lab-scale digester with variable headspace geometry, we studied the effect of headspace geometry on biogas and oxygen flow, biofilm growth, and microaeration efficiency. The experimental results from three tested headspace configurations were explained using a 3D computational fluid dynamics (CFD) model developed in COMSOL Multiphysics (COMSOL, Sweden). Our results show that adding properly designed baffles into the headspace can shift the biofilm growth to the baffles, and the walls of the digester can be protected. The model can be easily adjusted to other reactors only by modifying the geometry of its headspace. Therefore, the model can be a tool for ideal headspace design or evaluating the geometry effect on gas flow in existing reactors.

2. Materials and methods

2.1. Experimental study

2.1.1. Experimental setup

Two laboratory semi-continuous anaerobic digesters were built to study gas flow and sulfur-driven biofilm growth in the digester's headspace (Fig. 1). First digester with empty headspace served as control, and the second digester with variable geometry of the headspace was the tested digester (see Section 2.1.3). Both digesters were operated for 360 days (70 days of reaching steady state, 270 days of experiments) at the temperature of 44 °C. The chosen temperature was intended to maintain the conditions of the original inoculum. Both digesters were operated under microaerobic conditions, and with operational cycles. One operational cycle consisted of 50 min settling, 250 min stirring, and 2 min feeding. An artificial substrate: cheese whey solution (40 g/L) with the addition of K_2SO_4 (1.8 g/L) was used to increase the amount of H_2S in the biogas. The artificial substrate provided consistent conditions for all ongoing experiments, thus excluding any influence on the results due to variations in the composition of the dosed substrate. During the feeding time, the substrate was pumped into the bottom part of the reactor. Simultaneously, the supernatant from the top level of the liquid phase was pumped out from the reactor to keep the volume of the liquid phase constant. Both digesters had a total volume of 20 L with liquid and headspace volumes of 13 L and 7 L, respectively and were inoculated with anaerobic sludge from a mesophilic (44 °C) anaerobic digester (Česká Lípa, Czech Republic). Air is commonly used for microaeration purposes due to its simplicity and safety. However, air carries nitrogen contamination in the form of biogas, leading to the production of NOx during biogas combustion in a cogeneration unit (Rosenberg, personal communication, 30th May 2023), which is an undesirable air pollutant. Moreover, for our research, pure oxygen was advantageous, as it allowed us to dose oxygen more accurately and to quickly detect reactor contamination by air when measuring higher concentrations of N2 in the biogas, therefore, pure oxygen was used for microaeration. Oxygen was dosed continuously into the headspace of both reactors (4 mL h^{-1}) using a mass flow controller (Bronkhorst High-Tech B.V., Netherlands). A constant oxygen supply in this amount was chosen based on the need to monitor residual H_2S concentrations to assess the microaeration performance. Complete oxidation would not be desirable in this research.

2.1.2. Analytical methods

H_2S concentration in biogas was measured every 20 min by an online H_2S sensor (H_2S/SP-10000, Membrapor, Switzerland) equipped with an autosampler. The biogas production was measured by gas flow meter MilliGascounters (Dr.-Ing. RITTER Apparatebau GmbH & Co. KG, Germany) and was logged every 5 min. H_2S concentration, temperature, and biogas production measurement were controlled by an in-house program developed using LabVIEW 2012 software running on a CompactRIO system (National Instruments, US).

Chemical oxygen demand, total suspended solids and volatile suspended solids in liquid were measured twice a week according to the standard methods [14]. A gas chromatograph (GC 2010, Shimadzu, Japan) was used to measure volatile fatty acids in liquid. Samples preparation and the GC conditions set-up were done according to Andreides et al., 2021 [15]:

– Samples were centrifuged (13 000 rpm, 14 926 g, for 10 min)

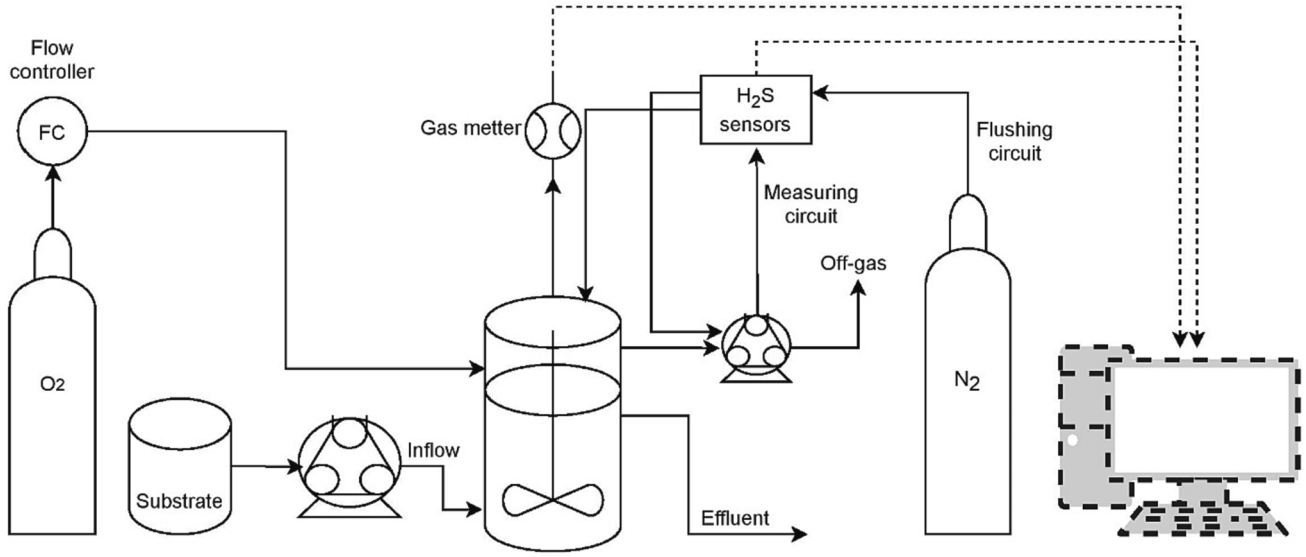


Fig. 1. Scheme of the laboratory set-up and H_2S measurement configuration.

- supernatants were filtered through a 0.45 μm syringe filter with a nylon membrane (AP-4549 T, Pall, USA).
- The gas chromatograph (GC 2010, Shimadzu, Japan) was equipped with a capillary column CP- Vax 58 (25 m length and 0.25 mm ID) and flame ionization detector.
- The oven column temperature was set at 134 °C.
- The injector port and detector temperature were 270 °C and 300 °C, respectively.
- The analysis time was 5.27 min.

The biogas and air composition in the reactors was measured twice a week. The conditions of the gas chromatograph GC-TCD (Shimadzu, Japan) were set as in Andreides et al., 2021 [16]:

- A stainless steel Hayesep D packed column (80/100) for biogas composition detection (CO_2, CH_4, air) and a 5A molecular sieve (80/100) for air composition detection (O_2, N_2) were used.
- The initial temperature of 150 °C, the detector temperature of 185 °C, and the isocratic analysis at 130 °C for both columns.
- Helium was used as the carrier gas.
- The analysis time for biogas and air detection was 3 and 6 min, respectively.

2.1.3. Experiments with variable headspace geometry

In the test digester, the cylindrical baffles (Fig. 2) were added to the digester to study the difference between biofilm growth positions in different headspace configurations. The baffles consisted of three concentric cylinders from laminated glass with 10, 15, and 20 cm diameters. The height of the cylinders was 6 cm. The diameter and height of the headspace were 24 cm and 10 cm, respectively. The oxygen inlet was placed on the digester's wall 3.5 cm under the ceiling. The cylindrical baffles supported biofilm growth and disrupted biogas flow, leading to the change of the dominant position of bacterial growth. The control digester's headspace was kept empty to imitate the usual conditions at the digesters.

The experiments were designed as follows:

Experiment 1: The headspace of both digesters (control and tested) was empty (no added cylinders).
Experiment 2: The headspace of the control digester was empty; the headspace of the tested digester had cylinders. The cylinders were suspended 3 cm above the liquid phase surface. Between the ceiling and cylinders was a small 1 cm gap.
Experiment 3: The headspace of the control digester was empty; the headspace of the tested digester had cylinders. The cylinders were in close contact with the liquid phase.

Before each experiment, the cylinders were immersed in the liquid phase of the reactor (anaerobic sludge) for one day to inoculate their surface by seeding bacteria. In all experiments, the biofilm had a growth period of at least 90 days. The experiment was terminated when no other biofilm shape and quality changes were visually observed. After each experiment, the biofilm was completely removed from the walls and cylinders.

2.2. Mathematical simulations

Software COMSOL Multiphysics with CFD module was used to study biogas flow in digesters. Only the headspace geometry was considered in the simulation to simplify the model. Based on the calculation of the Reynolds number at the reactor outlet, laminar biogas flow in the headspace was considered. The Mixture model/Laminar flow was used as a second physics to track the oxygen concentration in the modelled mixture (biogas/oxygen). Oxygen was modelled here as another phase with particle size taken as a diameter of oxygen molecule (0.36 nm). The gravity effect was modelled as well. Oxygen flux was set according to the actual dose set on the mass flow meter in the laboratory setup.

In the model, the following assumptions and calculations were taken into account:

- The liquid surface cone was considered negligible in headspace geometry.
- The surface rotation speed was derived according to the rotation cone's size. The calculated rotation speed of the liquid surface was 52.7 RPM (0.83067 s^{-1}), corresponding to 96 % of the stirrer speed (55 RPM).
- When the cylinders are in contact with liquid, the speed of the liquid surface rotation is not changed.

The experimentally measured biogas flow was considered to calculate the speed of the biogas coming from the liquid, and angular speed was calculated according to Eq. (1).

$$\omega^2 = \frac{2 \cdot g \cdot H}{r^2} \tag{1}$$

Where ω is an angular speed, g is the gravitational constant, H is a level rise during stirring, and r is a reactor radius. A cylindrical coordinate system was used for the calculations.

The speed of the liquid surface (*rpm*) was calculated according to Eq. (2).

$$rpm = \frac{\omega}{2\pi} \tag{2}$$

In the model, the 3D geometry of the reactor was built. As modelled gases, biogas and oxygen were used. Table 1 describes the properties of the gases related to the temperature of 44 °C.

The model was calculated with several meshes to ensure that results are not dependent on the mesh configuration. Ultimately, the mesh was set as automatic since no dependency was determined. The number of elements in the mesh was 785,220, with a maximum element size of 2.6 cm. The law of conservation of matter was also checked in the model.

2.3. Overall digesters performance and desulfurization efficiency

Both reactors were operated stably for 360 days. The values of pH and temperature were kept stable at 7.2–7.4 and 44 °C, respectively. Biogas composition was stable in both digesters (control and tested) for the whole operation period. After moisture content correction, biogas's average composition was 0.2 vol% of O_2, 60.0 vol% of CH_4, and 39.8 vol % of CO_2. Since the artificial substrate was used, the biogas composition was stable and the production was stably around 1200 mL per cycle.

During the microaeration of the reactors, the H_2S concentration fluctuated between 10 and a maximum of 160 ppm due to the discontinuous operation (50 min settling, 250 min stirring, 2 min feeding) (Fig. 3). The concentration of H_2S was intentionally kept at measurable levels to allow evaluation of the differences in microaeration efficiency. Nevertheless, the efficiency of microaeration remained above 97 % in all experiments. The H_2S peak period was observed at the beginning of each cycle. This phenomenon is caused by the stirring start and rapid release of H_2S from the liquid. SOB are not able to oxidize it immediately. Throughout the experiments, neither the shape of the curve nor the efficiency of microaeration changed. A detailed study of the influence of the stirring intensity on H_2S concentration is provided in Andreides et al. [3].

In Azizi et al. [17], one of the missing pieces of information about microaeration in literature is the time required to initiate complete desulfurization in the reactor. As written above, the biofilm was removed entirely from the reactor after every experiment. When there was no aeration, the H_2S concentration was around 6000 ppm. After the start of aeration, the concentration of H_2S dropped within one week, even though the biofilm was visibly growing for at least 60 days following the start of the oxygen dose. I.e., the biofilm area did not

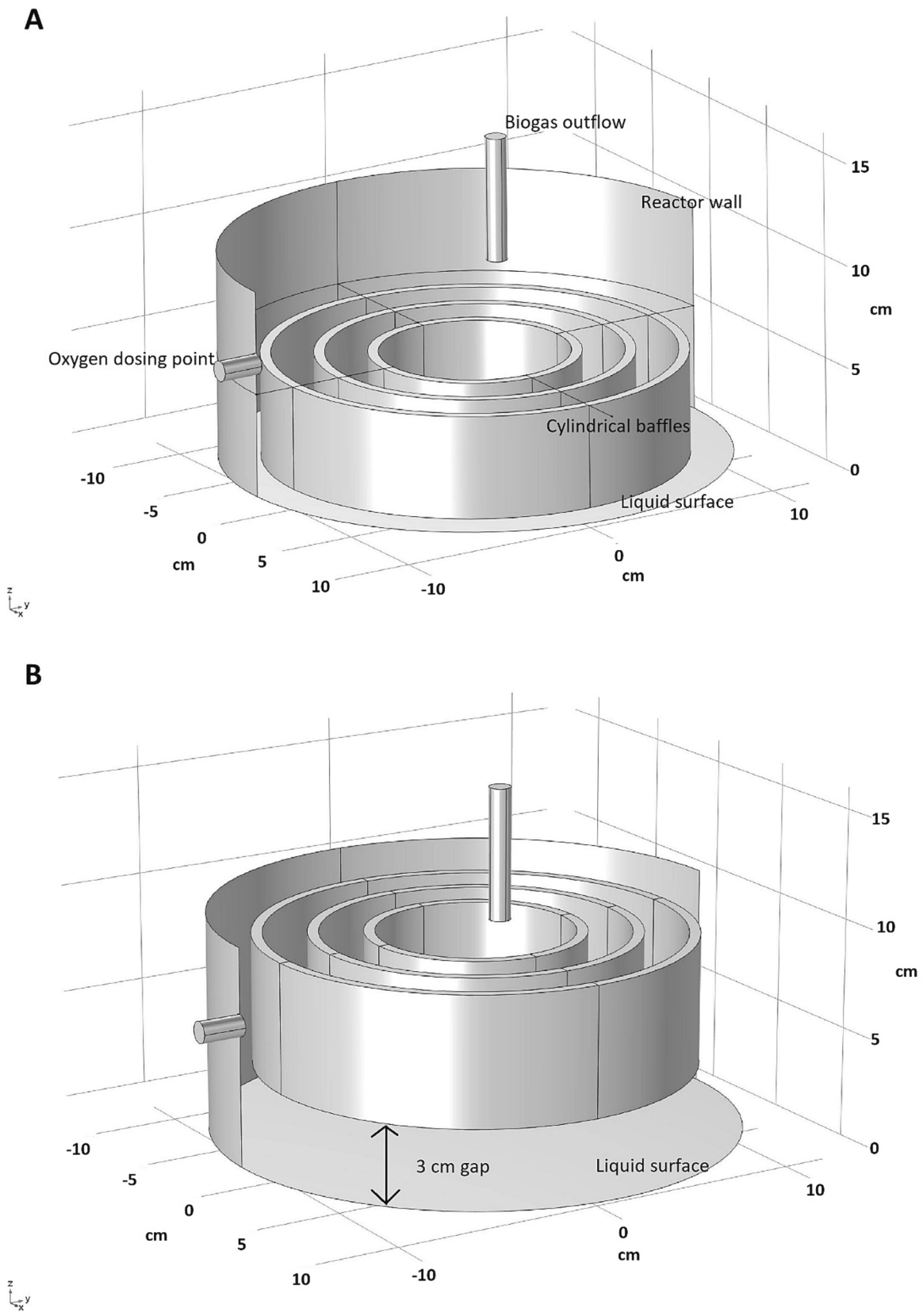


Fig. 2. Headspace of the tested digester with the cylinders installed serving as a surface for biofilm growth. The cylinders were suspended 3 cm above the surface (A) or immersed in the liquid phase (B).

Table 1
Properties of biogas and oxygen used in the mathematical model.

	Biogas	Oxygen
Density [kg/m^3]	1.2	1.23
Dynamic viscosity [Pa·s]	1.1125e^{-5}	2.1671e^{-5}

directly affect desulfurization efficiency.

2.4. Biofilm growth in different reactors configurations

During the experiment with empty headspace (Experiment 2), the biofilm covered the entire digester wall, including the ceiling (Fig. 4A). Such a phenomenon is usually visible in full-scale applications.

When the cylinders had no connection with the liquid phase (Experiment 2), biofilm covered the digester's walls up to 3 cm from the liquid surface (Fig. 4B). There was no biofilm between the wall and the outer cylinder. The sulfur deposition at the cylinders was almost not observed.

During the experiment with cylinders in contact with the liquid, the biofilm covered cylinders only, and the digester itself was clean (Fig. 4C). Fig. 5 shows the view of the cylinders from the underside that was in contact with the liquid. The structure of the biofilm was visually similar to the biofilm on the wall in the second experiment. The results were compared with simulation results and are discussed in Section 3.3.

2.5. Simulation of biogas flow pattern in the headspace

2.5.1. Oxygen flow

Based on the simulation results, oxygen was in all three experiments spun at the point of oxygen entry after the first contact with biogas (Fig. 6). After entering the reactor, oxygen was immediately carried to the biogas outlet. Some oxygen was washed out from the reactor with biogas, explaining the oxygen presence (0.2 % vol.) in the biogas. The remaining oxygen travelled through the center of the headspace to the liquid surface, where it picked up speed again.

The position of the oxygen dosing point was not ideal. The amount of oxygen required for complete H_2S oxidation was higher due to the washout from the reactor, so the cost of the required oxygen would increase at full-scale. In literature, the stochiometric ratio O_2/S^{2-} required for complete desulfurization varies among different studies (E.g. 0.5 [11], 1 – 1.4 [9], 1 [16], or 5.5 [18]. Based on the simulation results, when the oxygen is used for desulfurization only, the difference might be caused by the position of the dosing point and the subsequent ratio between oxygen available to sulfide oxidizing bacteria and oxygen drained from the reactor with biogas. However, the evaluation is impossible since there is no precise information about the position of the dosing point and geometry of the headspace in previously published papers.

2.5.2. The flow of a biogas/oxygen mixture

In the experiment with empty headspace, the biogas coming from the liquid was carried to the walls by centrifugal force together with oxygen (Fig. 6A). The gas mixture was evenly distributed throughout the headspace. The conditions for sulfide oxidizing bacteria were not significantly prioritized at specific places; therefore, the biofilm covered the outer wall of the whole headspace.

In the experiment with cylinders suspended 3 cm above the liquid (Exp.2) (6B), the mixture under the cylinders behaved similarly to the experiment with empty headspace. Once the mixture reached the cylinders, the mixture flowed to the ceiling and continued to the gaps between the cylinders. The mixture flowed rapidly from top to bottom between the cylinders and returned to the liquid down the center (Fig. 7A). Assuming only the z-component of the velocity describing the vertical speed of the mixture (Fig. 8), the mixture flowed between the cylinders only in a downward direction, shortening the residence time between the gaps. The sulfur deposition was observed mainly at the reactor's wall under the cylinders, which was most likely caused by this mixture circulation, limited residence time between the cylinders and the highest concentration of H_2S evolving from the liquid and pushed to the wall by centrifugal force.

In the experiment with cylinders in direct contact with liquid (Exp. 3), the gas mixture could not be directly pushed to the walls since the cylinders interrupted the flow (6C). The mixture flowed from the center from one gap to another. It reached the reactor's wall at the end (Fig. 7B). The residence time of the mixture between the cylinders could have caused all oxygen was entirely consumed, resulting in no oxidation possibility in the gap between the last cylinder and the wall. Assuming only the z-component of the velocity, the vertical speed decreased twice compared to the experiment with cylinders suspended 3 cm above the liquid. Fig. 8 proves that the mixture flowed up and down from one gap to another. In the experiment with cylinders in direct contact with liquid, the biofilm growth was apparently prioritized between the cylinders, which correlates to the experimental results (Figs. 4 and 5). The simulation results correlate with the observed sulfur deposition in the reactor in all three experiments.

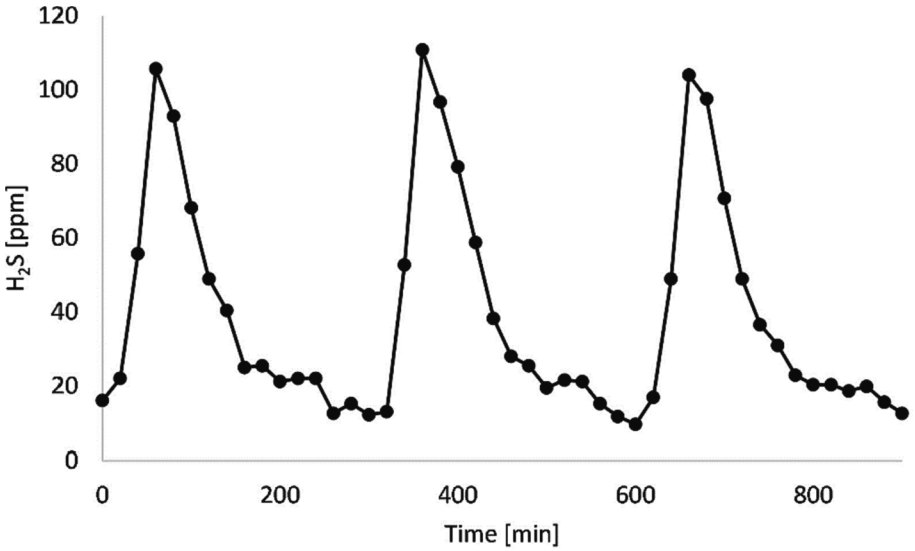


Fig. 3. Typical curve of the H_2S concentration in the headspace of SBR (three randomly chosen cycles).

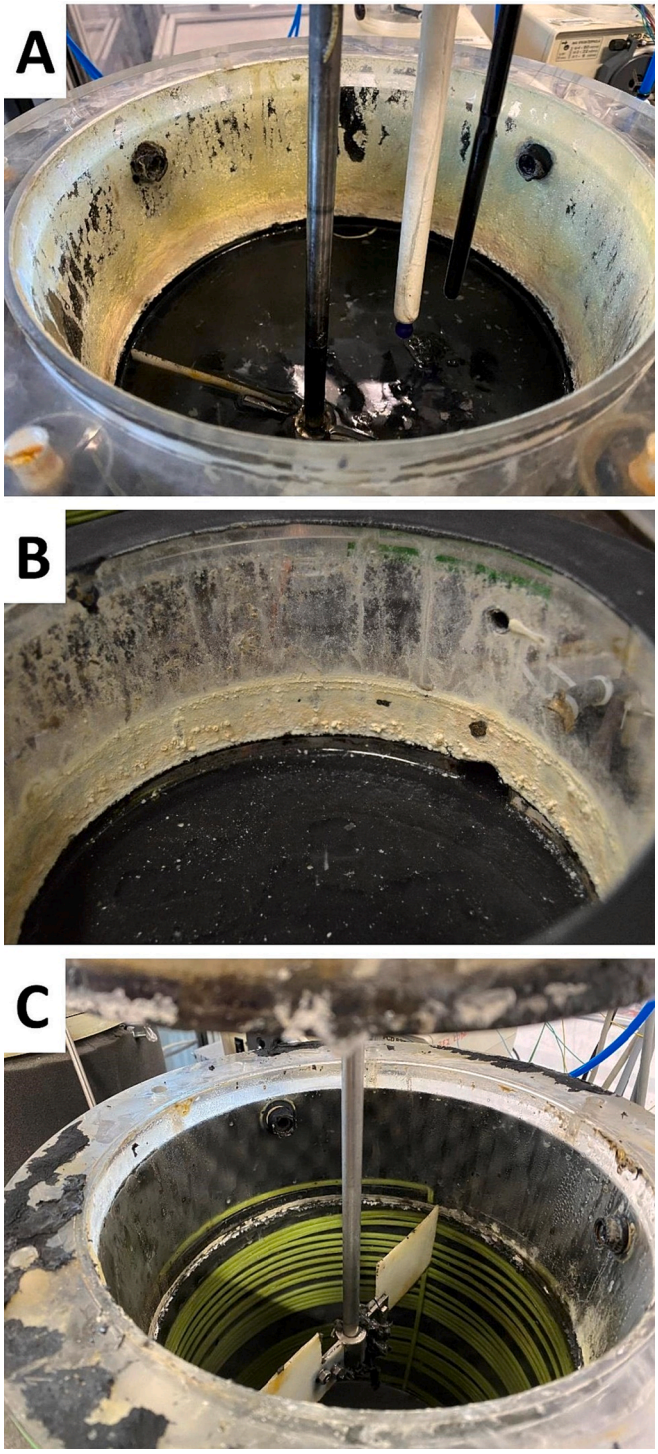


Fig. 4. A view into the open reactor in each experiment; Biofilm growing on the wall of the digester's headspace in the experiment with empty headspace (A), the cylinders suspended 3 cm above the liquid (B), and the cylinders in direct contact with liquid (C).

2.6. Implications for reactor design

According to the results of the mathematical simulation, the headspace geometry had a crucial effect on sulfur deposition position and oxygen requirements. To the best of our knowledge, this is the first time the complex headspace geometry effect on microaeration was evaluated.

Adding the cylinders into the headspace had no adverse effect on microaeration efficiency; only the biofilm position was affected. From the technological point of view, the worst scenario was the empty headspace, where all the walls and structures in the headspace were completely covered with sulfur-driven biofilm and could be degraded by corrosion. On the other hand, when the cylinders were in direct contact with liquid, the microaeration efficiency was not affected, and the digester's wall was efficiently protected.



Fig. 5. Cylinders covered with SOB biofilm after the experiment with cylinders in direct contact with liquid; View of the cylinders from the underside that was in contact with the liquid.

The CFD model developed can be used in every bioreactor modelling where the gas is evolved into the headspace from rotating liquid. Given that we used an artificial substrate, the rate of biogas emerging from the liquid to the gas phase was constant. If biogas production is diverse, it is necessary to consider that this rate may influence the mixing patterns in headspace and conduct scenario analyses where several biogas flow rates are investigated for a specific reactor.

Installing the baffles into the fermenter will increase investment costs compared to classic microaeration, especially when the fermenter has already been built. On the other hand, the baffles installed will increase the technology's lifetime and help to remove sulfur biofilm effectively during the cleaning period. Compared to investing in the new ex-situ desulphurization, the costs will be many times lower, and no additional area will be required. The final financial savings will depend on the structure of the existing or newly installed fermenter, the material used and the shape of the installation and need to be evaluated in dedicated full-scale experiments.

2.7. Further research

In the research, an artificial substrate was used to achieve the same steady state in all experiments. There is a presumption that results from biofilm growth would be similar when the industrial substrate was used since the substrate for bacteria is not related to the composition of the liquid but H_2S presence in biogas. However, this needs to be confirmed.

In this study, sophisticated and accurate methods for online H_2S and biogas monitoring were used, which allowed obtaining strong dataset about the behavior of the gases in the experiments. On the other hand, detailed sulfur analyses of the biofilm were not done due to the lack of biological material (dried biofilm) for these analyses. Results from the biofilm composition and the microbial abundance in each experiment would support the CFD model results.

The model presented in this study showed the ability to describe gas flow in different headspace configurations only by changing the geometry of the headspace in the model. The next step of the research would be adding biological processes into the model and simulating the biofilm growth and the H_2S/oxygen consumption in space. Such simulation

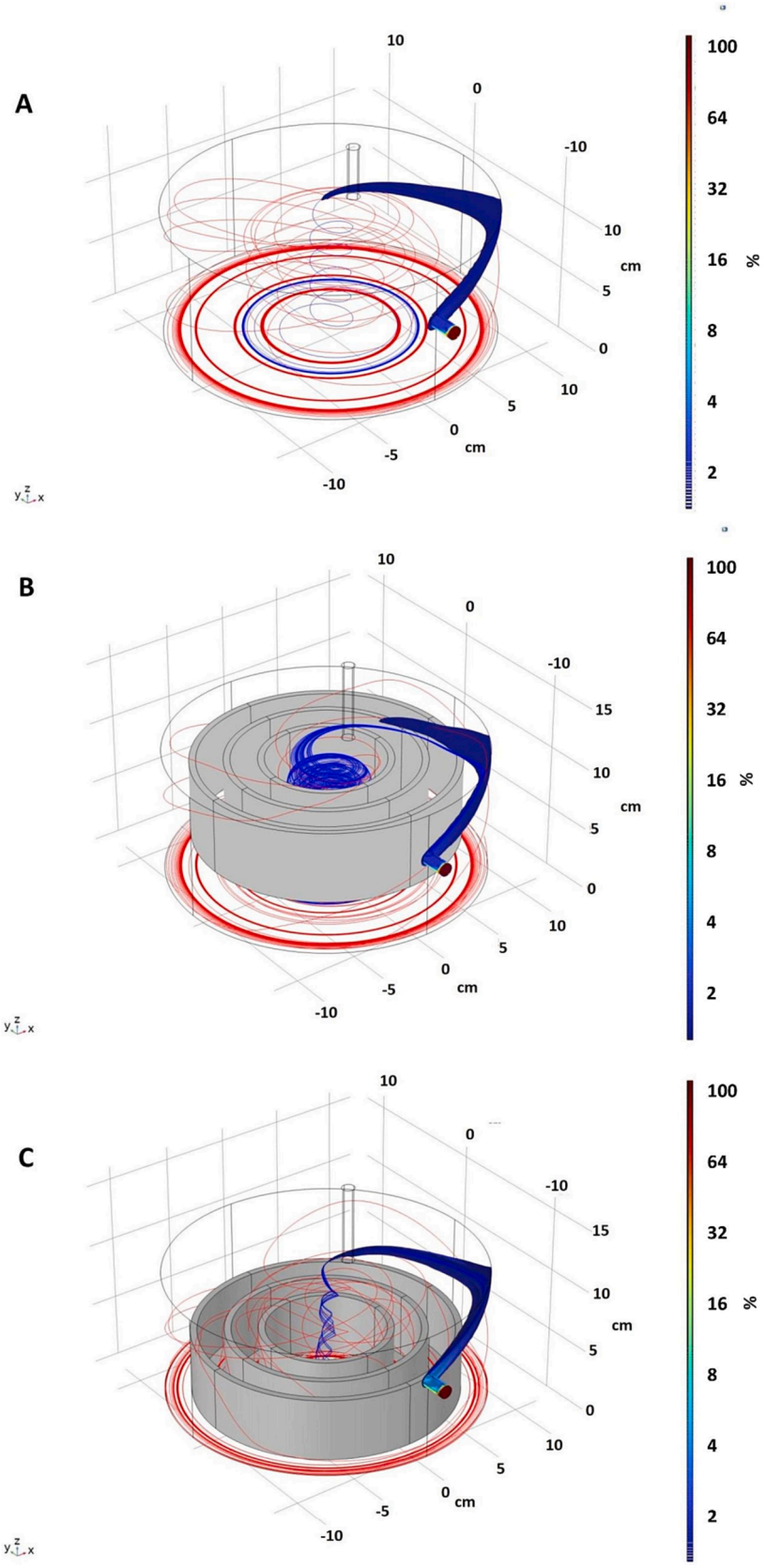


Fig. 6. Flow of biogas (red)/oxygen (blue) mixture in the experiment with empty headspace (A), the cylinders suspended 3 cm above the liquid (B), and the cylinders in direct contact with liquid (C). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

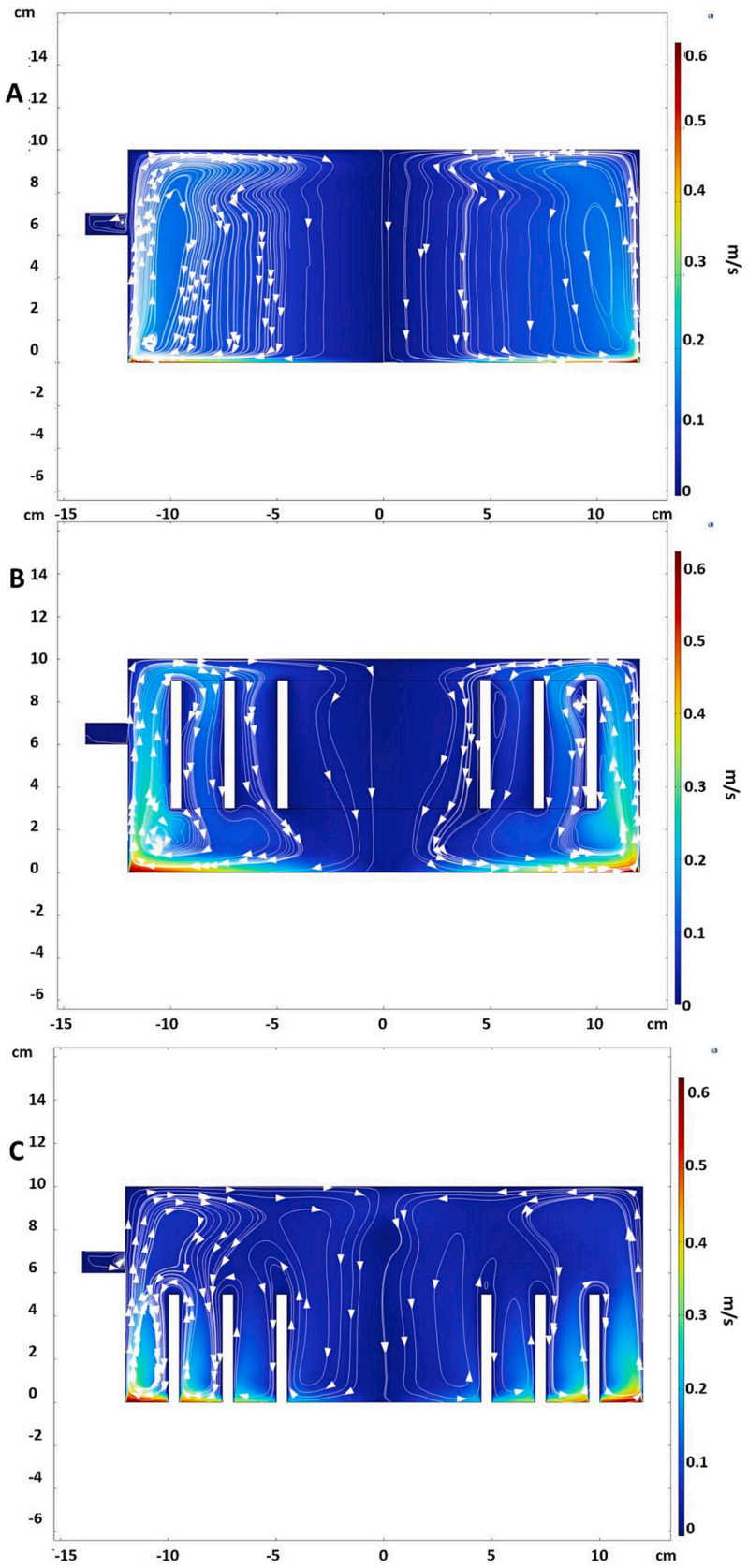


Fig. 7. Cross-section of reactor's headspace showing biogas/oxygen mixture flow in the experiment with empty headspace (A), the cylinders suspended 3 cm above the liquid (B), and the cylinders in direct contact with liquid (C).

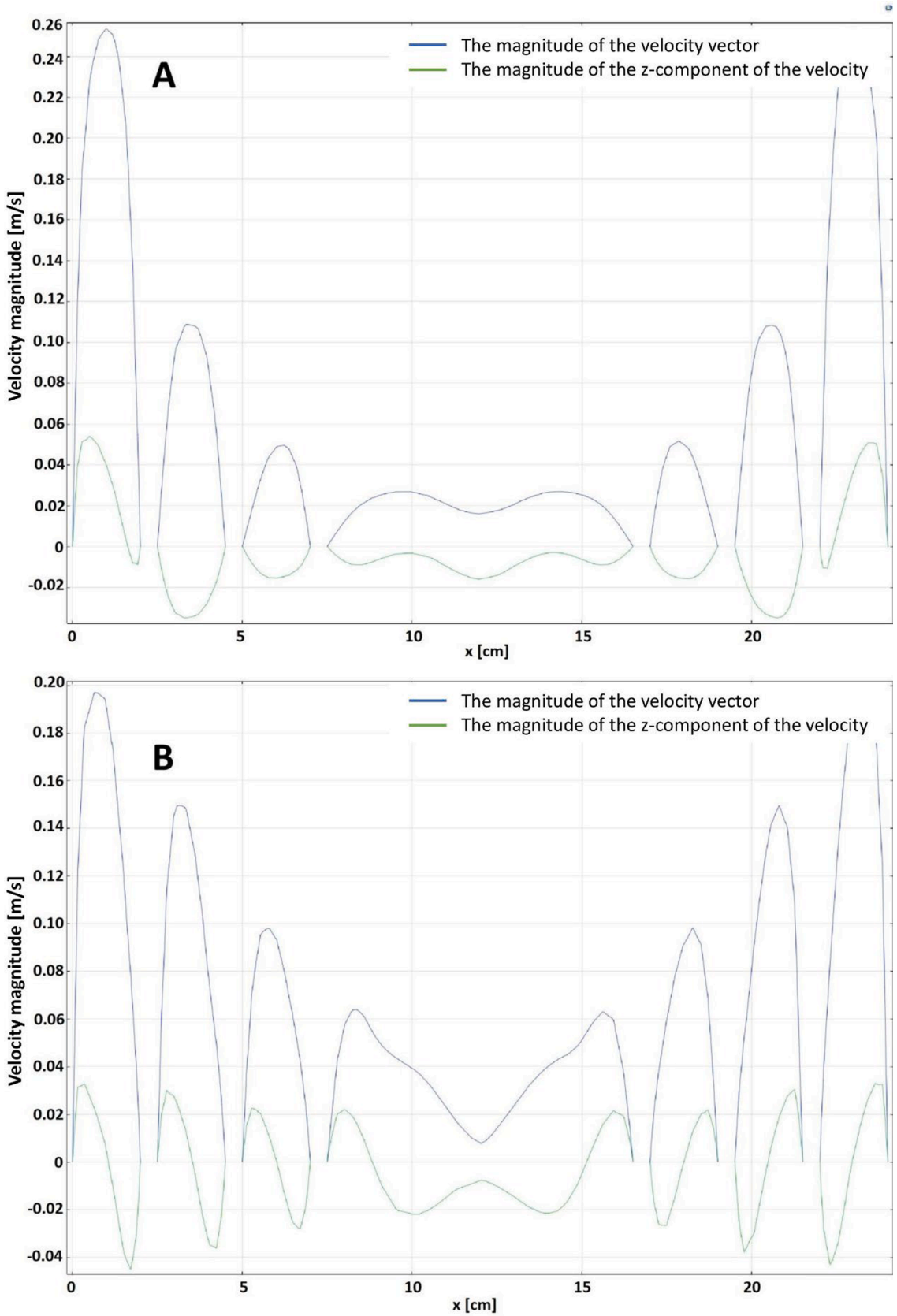


Fig. 8. Comparison of the magnitude of the velocity vector and z-component of the velocity in the cylinders gaps at the height of 4 cm above the liquid surface in the experiment with the cylinders suspended 3 cm above the liquid (A), and the cylinders in direct contact with liquid (B).

requires high computational complexity [19]. Adding a biological process to the model could also provide assurance that there will be no development of SOB on undesirable surfaces in the long term. In this study, we terminated all experiments only when, for several weeks, there was no visual evidence of culture development. Simulation results also suggested that there was no biological activity on the reactor walls due to substrate scarcity. However, a long-term simulation would bring even greater certainty in fermenter design.

The mathematical model was developed based on knowledge of fluid flow. Validation of the model involved comparing the simulation results with experimental results, where gas flow, and consequently the accessibility of the substrate to bacteria, corresponded to actual increases in biofilm observed in the experiments. In the next phase of the research, experiments, including simulations, should be replicated in a full-scale situation.

3. Conclusions

The CFD model developed in this study described the flow of the gases above the rotating liquid surface in anaerobic digesters with different reactor headspace configurations. The model supported by experimental data explained the effect of headspace geometry on biogas and oxygen flow patterns, microaeration efficiency, and the sulfur deposition in the headspace. We also showed that when internal baffles placed in the headspace disturbed biogas flow, the position of biofilm growth was altered. The fact that biogas rotated in the headspace directly influenced which surfaces were in contact with both H_2S and oxygen and hence defined where biofilm developed. Finally, the amount of oxygen required for efficient desulfurization was shown to depend on the position of the oxygen dosing point. Bioreactor designers can use these findings to design headspace geometry so biofilm will form on predefined surfaces while protecting the reactor's walls from biofilm-induced corrosion. The research hypothesis that the headspace geometry modification can influence oxygen consumption, biogas flow and subsequent sulfur deposition in the headspace was confirmed.

CRediT authorship contribution statement

Markéta Andreides: Writing – original draft. **Lucie Pokorná-Krayzelová:** Supervision. **Jan Bartáček:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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